

Week B.

Prob 1. Prove hash family $\mathcal{H} = \{h_{a,b} \mid h_{a,b}(x) = a \cdot x + b, a \in \{0,1\}^k, b \in \{0,1\}\}$ is pairwise independent.

For arbitrary $x_1 \neq x_2 \in \{0,1\}^k, y_1 \neq y_2 \in \{0,1\}$. We prove

$$P_{h \in \mathcal{H}}(h(x_1) = y_1 \wedge h(x_2) = y_2) = \frac{1}{4}.$$

Suppose $x_1' \neq x_2'$ without loss of generality. Let $z_1 = \sum_{i=1}^{y_1+k} x_1' a_i, z_2 = \sum_{i=1}^{y_2+k} x_2' a_i$, modulo 2. Thus $h(x_1) = y_1$ iff $x_1' a' + b = z_1, h(x_2) = y_2$ iff $x_2' a' + b = z_2$.

So $h(x_1) = y_1 \wedge h(x_2) = y_2$ iff $a' = 0, b = z_1$ (when $z_1 = z_2$)
 $a' = 1, b = x_1' + z_1$ (when $z_1 \neq z_2$).

Since (a', b) has only 1 valid in 4 cases. the prob. is $\frac{1}{4}$.

Prob 2.

a. Consider random walk $X_0, X_1, \dots$ on $[0, n] \cap \mathbb{N}$, every branch equally chosen. Let T_i be $E(\text{steps taken to } n \text{ start from } i)$. Prove $T_i \leq n^2$.

Let $F_i = E(\text{steps taken to } i \text{ start from } i-1)$. then $T_i = \sum_{j=i+1}^n F_j$.

We have $F_i = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot (1 + \bar{F}_{i-1} + \bar{F}_i) = 1 + \frac{1}{2}(\bar{F}_{i-1} + \bar{F}_i)$.
 then $F_i = 2 + \bar{F}_{i-1}$.

Since $F_1 = 1$. We have $\bar{F}_i = 2i - 1$. then $T_i = \sum_{j=i+1}^n 2j - 1 = n^2 - i^2$.

Obviously $T_i \leq n^2$. (In fact, this method can solve random walk on tree).

b. Consider a random algo. for 2-SAT: ① Choose an arbitrary initial assignment.
 ② Choose an ^{arbitrary} clause not satisfied. Equally randomly pick one side of two to flip.
 Use Markov inequality to show the algo has error prob. $\frac{1}{2}$. ③ If ② did $2n^2 - 1$ times and not right, reject. otherwise accept.

If the instance has no solution, the algo will reject. Suppose a solution is $x \in \{0,1\}^n$. For a assignment y , let $\|y\| = \sum_{i=1}^n [y_i \neq x_i]$. then $0 \leq \|y\| \leq n$. if $\|y\| = 0$, y is solution.

Suppose for a assignment $\|y\| > 0$, we find a unsatisfied clause involving variable p_i, p_j . then we must have $y_i \neq x_i$ or $y_j \neq x_j$. otherwise the clause is satisfied. We randomly pick p_i or p_j . ~~then sample~~ $\{i, j\}$ each $\frac{1}{2}$. then ~~$P(y_i \neq x_i)$~~ , $P(y_i \neq x_i) \geq \frac{1}{2}$.

If $y_i \neq x_i$, then after flip y_i , called y' , we have $\|y'\| = \|y\| - 1$.
 if $y_i = x_i$, $\|y'\| = \|y\| + 1$.



Initially. $\|y_0\| \leq n$. if $\|y_0\| = n$. then $\|y_1\|$ must be $n-1$. So

$X_0 = \|y_0\|$, $X_1 = \|y_1\|$, $\dots$ is a random walk on $[0, n]$ with decrease
So $E(\text{step taken to reach } 0) \leq n^2$. $\text{prob.} \geq \frac{1}{2}$.

Thus $P(\text{step} \geq 2n^2) \leq \frac{n^2}{2n^2} = \frac{1}{2}$, that is the algo has $\frac{1}{2}$ or more prob.
to be correct. and gives a solution.

